

Oral 1: Clinical Translation I – Medical Image Computing

Monday, October 9, 09:00 to 10:30

Exhibit Hall A – Main Hall

Session Chairs: Charles Kahn, University of Pennsylvania, USA
Curtis Langlotz, Stanford University, USA

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| 09:00-09:30 | From Algorithm to the Clinic: Critical issues to Consider during Translation of Image Analysis Advances into Daily Clinical Practice
Invited Session Speaker: Mariam Aboian, Yale School of Medicine, USA |
| 09:30-09:45 | Shifting More Attention to Breast Lesion Segmentation in Ultrasound Videos
Speaker: Huazhu Fu, A*STAR, Singapore |
| 09:45-10:00 | Foundation Ark: Accruing and Reusing Knowledge for Superior and Robust Performance
Speaker: DongAo Ma, Arizona State University, USA |
| 10:00-10:15 | CheXstray: A Real-Time Multi-Modal Monitoring Workflow for Medical Imaging AI
Speaker: Jameson Merkow, Microsoft, USA |
| 10:15-10:30 | Thinking Like Sonographers: A Deep CNN Model for Diagnosing Gout from Musculoskeletal Ultrasound
Speaker: Zhi Cao, Nanjing University of Aeronautics and Astronautics, China |

Oral 2: Computational Pathology

Monday, October 9, 09:00 to 10:30

Ballroom A – Parallel Hall

Session Chairs:

Saad Nadeem, Memorial Sloan Kettering Cancer Center, USA

Daniel Racoceanu, Sorbonne University, France

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| 09:00-09:15 | Pathology-and-genomics Multimodal Transformer for Survival Outcome Prediction
Speaker: Kexin Ding, University of North Carolina at Charlotte, USA |
| 09:15-09:30 | NASDM: Nuclei-Aware Semantic Histopathology Image Generation Using Diffusion Models
Speaker: Aman Shrivastava, University of Virginia, USA |
| 09:30-09:45 | DAS-MIL: Distilling Across Scales for MIL Classification of Histological WSIs
Speaker: Gianpaolo Bontempo, University of Modena and Reggio Emilia, Italy |
| 09:45-10:00 | MulHiST: Multiple Histological Staining for Thick Biological Samples via Unsupervised Image-to-Image Translation
Speaker: Lulin Shi, Hong Kong University of Science and Technology, China |
| 10:00-10:15 | Multi-task Learning of Histology and Molecular Markers for Classifying Diffuse Glioma
Speaker: Xiaofei Wang, University of Cambridge, United Kingdom |
| 10:15-10:30 | Gene-induced Multimodal Pre-training for Image-omic Classification
Speaker: Ting Jin, East China Normal University, China |

Oral 3: Machine Learning I – Semi-Supervised & Self-Supervised

Monday, October 9, 14:30 to 16:00

Exhibit Hall A – Main Hall

Session Chairs:

Shekoofeh Azizi, Google, USA

Davood Karimi, Harvard University, USA

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| 14:30-14:45 | Multi-modal Variational Autoencoders for normative modelling across multiple imaging modalities
Speaker: Ana Lawry Aguila, University College London, United Kingdom |
| 14:45-15:00 | Towards AI-driven radiology education: A self-supervised segmentation-based framework for high-precision medical image editing
Speaker: Kazuma Kobayashi, National Cancer Center Research Institute, Japan |
| 15:00-15:15 | Prompt-MIL: Boosting Multi-Instance Learning Schemes via Task-specific Prompt Tuning
Speaker: Jingwei Zhang, Stony Brook Universtiy, USA |
| 15:15-15:30 | LOTUS: Learning to Optimize Task-based US representations
Speaker: Yordanka Velikova, Technical University of Munich, Germany |
| 15:30-15:45 | Category-level Regularized Unlabeled-to-labeled Learning for Semi-supervised Prostate Segmentation with Multi-site Unlabeled Data
Speaker: Zhe Xu, The Chinese University of Hong Kong, Hong Kong SAR, China |
| 15:45-16:00 | Correlation-Aware Mutual Learning for Semi-supervised Medical Image Segmentation
Speaker: Shengbo Gao, Deepwise AI Lab, China |

Oral 4: Computer Assisted Interventions and Surgery

Monday, October 9, 14:30 to 16:00

Ballroom A – Parallel Hall

Session Chairs:

Sophia Bano, University College London, UK

Mathias Unberath, Johns Hopkins University, USA

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| 14:30-14:45 | Detecting the Sensing Area of A Laparoscopic Probe in Minimally Invasive Cancer Surgery
Speaker: Baoru Huang, Imperial College London, United Kingdom |
| 14:45-15:00 | FLIm-based In Vivo Classification of Residual Cancer in the Surgical Cavity during Transoral Robotic Surgery
Speaker: Mohamed Hassan, University of California Davis, USA |
| 15:00-15:15 | FocalErrorNet: Uncertainty-aware focal modulation network for inter-modal registration error estimation in ultrasound-guided neurosurgery
Speaker: Soorena Salari, Concordia University, Canada |
| 15:15-15:30 | From Tissue to Sound: Model-based Sonification of Medical Imaging
Speaker: Sasan Matinfar, Technical University of Munich, Germany |
| 15:30-15:45 | ConTrack: Contextual Transformer for Device Tracking in X-ray
Speaker: Yue Zhang, Siemens Healthineers, USA |
| 15:45-16:00 | A Transfer Learning Approach to Localise a Deep Brain Stimulation Target
Speaker: Ying-Qiu Zheng, University of Oxford, United Kingdom |

Oral 5: Machine Learning II – Towards Transparent AI

Tuesday, October 10, 08:00 to 09:30

Exhibit Hall A – Main Hall

Session Chairs:

Ruogu Fang, University of Florida, USA

Zongyuan Ge, Monash University, Australia

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| 08:00-08:15 | Interpretable Medical Image Classification using Prototype Learning and Privileged Information
Speaker: Luisa Gallée, Experimental Radiology, University Hospital Ulm, Germany |
| 08:15-08:30 | How Reliable are the Metrics Used for Assessing Reliability in Medical Imaging?
Speaker: Mayank Gupta, Indian Institute of Technology Delhi, India |
| 08:30-08:45 | Interpretable Deep Biomarker for Serial Monitoring of Carotid Atherosclerosis Based on Three-Dimensional Ultrasound Imaging
Speaker: Xueli Chen, City University of Hong Kong, Hong Kong SAR, China |
| 08:45-9:00 | B-Cos Aligned Transformers Learn Human-Interpretable Features
Speaker: Manuel Tran, Technical University Munich, Germany |
| 09:00-09:15 | An Explainable Geometric-Weighted Graph Attention Network for Identifying Functional Networks Associated with Gait Impairment
Speaker: Favour Nerrise, Stanford University, USA |
| 09:15-09:30 | A Reliable and Interpretable Framework of Multi-view Learning for Liver Fibrosis Staging
Speaker: Zheyao Gao, Fudan University, China |

Oral 6: Neuroimaging – Morphology to Functionality

Tuesday, October 10, 08:00 to 09:30

Ballroom A – Parallel Hall

Session Chairs:

Minjeong Kim, University of North Carolina at Greensboro, USA

Matthias Wilms, University of Calgary, Canada

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| 08:00-08:15 | Dynamic Functional Connectome Harmonics
Speaker: Hoyt Patrick Taylor, University of North Carolina at Chapel Hill, USA |
| 08:15-08:30 | Mixing Temporal Graphs with MLP for Longitudinal Brain Connectome Analysis
Speaker: Hyuna Cho, Pohang University of Science and Technology, South Korea |
| 08:30-08:45 | Unified surface and volumetric inference on functional imaging data
Speaker: Thomas F. Kirk, University of Nottingham, United Kingdom |
| 08:45-9:00 | Multi-task Joint Prediction of Infant Cortical Morphological and Cognitive Development
Speaker: Xinrui Yuan, The University of North Carolina at Chapel Hill, USA |
| 09:00-09:15 | Flexible Unfolding of Circular Structures for Rendering Textbook-Style Cerebrovascular Maps
Speaker: Leonhard Rist, Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany |
| 09:15-09:30 | Bidirectional Mapping with Contrastive Learning on Multimodal Neuroimaging Data
Speaker: Kai Ye, University of Pittsburgh, USA |

Oral 7: Computer Aided Diagnosis with Longitudinal and Multi-modal Data

Tuesday, October 10, 16:30 to 18:00

Exhibit Hall A – Main Hall

Session Chairs:

Samuel Kadoury, Polytechnique Montréal, Canada

Mirabela Rusu, Stanford University, USA

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| 16:30-16:45 | Longitudinal Multimodal Transformer Integrating Imaging and Latent Clinical Signatures From Routine EHRs for Pulmonary Nodule Classification
Speaker: Thomas Li, Vanderbilt University, USA |
| 16:45-17:00 | Graph-theoretic automatic lesion tracking and detection of patterns of lesion changes in longitudinal CT studies
Speaker: Leo Joskowicz, Hebrew University of Jerusalem, Israel |
| 17:00-17:15 | Utilizing Longitudinal Chest X-Rays and Reports to Pre-Fill Radiology Reports
Speaker: Qingqing Zhu, National Institutes of Health, USA |
| 17:15-17:30 | Improving Outcome Prediction of Pulmonary Embolism by De-Biased Multi-Modality Model
Speaker: Shreyas Kulkarni, Brown University, USA |
| 17:30-17:45 | Contrastive Masked Image-Text Modeling for Medical Visual Representation Learning
Speaker: Cheng Chen, Massachusetts General Hospital and Harvard Medical School, USA |
| 17:45-18:00 | Multimodal Deep Fusion in Hyperbolic Space for Mild Cognitive Impairment Study
Speaker: Dajiang Zhu, The University of Texas at Arlington, USA |

Oral 8: Surgical Visualization and Data Science

Tuesday, October 10, 16:30 to 18:00

Ballroom A – Parallel Hall

Session Chairs:

Sandy Engelhardt, Heidelberg University Hospital, Germany

Mobarakol Islam, University College London, UK

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| 16:30-16:45 | Self-supervised Sim-to-Real Kinematics Reconstruction for Video-based Assessment of Intraoperative Suturing Skills
Zijun Cui, University of Southern California, USA |
| 16:45-17:00 | POV-Surgery: A Dataset for Egocentric Hand and Tool Pose Estimation During Surgical Activities
Speaker: Rui Wang, ETH Zürich, Switzerland |
| 17:00-17:15 | Intelligent Virtual B-scan Mirror (IVBM)
Speaker: Michael Sommersperger, Technical University of Munich, Germany |
| 17:15-17:30 | EndoSurf: Neural Surface Reconstruction of Deformable Tissues with Stereo Endoscope Videos
Speaker: Xuelian Cheng, Monash University, Australia |
| 17:30-17:45 | Neural LerPlane Representations for Fast 4D Reconstruction of Deformable Tissues
Speaker: Chen Yang, Shanghai Jiao Tong University, China |
| 17:45-18:00 | ACT-Net: Anchor-context Action Detection in Surgery Videos
Speaker: Jiang Liu, University of Birmingham, United Kingdom |

Oral 9: Segmentation – Methods and Applications

Wednesday, October 11, 08:00 to 09:30

Exhibit Hall A – Main Hall

Session Chairs:

Ulas Bagci, Northwestern University, USA

Herve Lombaert, ETS Montreal, Canada and Inria, France

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| 08:00-08:15 | SwinMM: Masked Multi-view with Swin Transformers for 3D Medical Image Segmentation
Speaker: Yiqing Wang, Shanghai Jiao Tong University, China |
| 08:15-08:30 | MultiTalent: A Multi-Dataset Approach to Medical Image Segmentation
Speaker: Constantin Ulrich, German Cancer Research Center, Germany |
| 08:30-08:45 | Structure-Preserving Instance Segmentation via Skeleton-Aware Distance Transform
Speaker: Donglai Wei, Boston College, USA |
| 08:45-09:00 | Pelvic Fracture Segmentation Using a Multi-scale Distance-weighted Neural Network
Speaker: Yanzhen Liu, Beihang University, China |
| 09:00-09:15 | Robust and Generalisable Segmentation of Subtle Epilepsy-causing Lesions: a Graph Convolutional Approach
Speaker: Hannah Spitzer, LMU University Hospital, Germany |
| 09:15-09:30 | CorSegRec: A Topology-Preserving Scheme for Extracting Fully-Connected Coronary Arteries from CT Angiography
Speaker: Dinggang Shen, ShanghaiTech University & Shanghai United Imaging Intelligence Co., Ltd., China |

Oral 10: Clinical Translation II – Computer Assisted Intervention

Wednesday, October 11, 08:00 to 09:30

Ballroom A – Parallel Hall

Session Chairs:

Masaru Ishii, Johns Hopkins Hospital, USA

Daniel Hashimoto, University of Pennsylvania, USA

08:00-08:30

Panel Discussion

Panelists:

Mariam Aboian, Yale School of Medicine, USA

Daniel Hashimoto, University of Pennsylvania, USA

Russell Taylor, Johns Hopkins University, USA

Sandrine de Ribaupierre, Western University, Canada

08:30-08:45

From Mesh Completion to AI Designed Crown

Speaker: Golriz Hosseinimanes, Polytechnique Montreal, Canada

08:45-09:00

Optical Ultrasound Imaging for Endovascular Repair of Abdominal Aortic Aneurysms: A Pilot Study

Speaker: Adrien Desjardins, University College London, United Kingdom

09:00-09:15

Estimated time to surgical procedure completion: An exploration of video analysis methods

Speaker: Yariv Colbeci, Theodor, Israel

09:15-09:30

Automatic Surgical Reconstruction for Orbital Blow-out Fracture via Symmetric Prior Anatomical Knowledge-Guided Adversarial Generative Network

Speaker: Jiangchang Xu, Shanghai Jiao Tong University, China

Oral 11: Machine Learning III – Advances in Learning Strategies

Wednesday, October 11, 13:00 to 14:30

Exhibit Hall A – Main Hall

Session Chairs:

Bernhard Kainz, Imperial College London, UK and FAU Erlangen-Nürnberg, Germany

Xiaoxiao Li, University of British Columbia, Canada

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| 13:00-13:15 | Uncertainty and Shape-Aware Continual Test-Time Adaptation for Cross-Domain Segmentation of Medical Images
Speaker: Jiayi Zhu, University of New South Wales, Australia |
| 13:15-13:30 | Open-Ended Medical Visual Question Answering Through Prefix Tuning of Language Models
Speaker: Tom van Sonsbeek, University of Amsterdam, the Netherlands |
| 13:30-13:45 | Joint prediction of response to therapy, molecular traits, and spatial organisation in colorectal cancer biopsies
Speaker: Ruby Wood, University of Oxford, United Kingdom |
| 13:45-14:00 | Speech Audio Synthesis from Tagged MRI and Non-Negative Matrix Factorization via Plastic Transformer
Speaker: Xiaofeng Liu, Harvard Medical School, USA |
| 14:00-14:15 | Deployment of Image Analysis Algorithms under Prevalence Shifts
Speakers: Patrick Godau and Piotr Kalinowski, German Cancer Research Center (DKFZ), Germany |
| 14:15-14:30 | ProtoASNet: Dynamic Prototypes for Inherently Interpretable and Uncertainty-Aware Aortic Stenosis Classification in Echocardiography
Speaker: Hooman Vaseli, University of British Columbia, Canada |

Oral 12: Physics-based Image Formation and Reconstruction

Wednesday, October 11, 13:00 to 14:30

Ballroom A – Parallel Hall

Session Chairs:

Angelica Aviles-Rivero, University of Cambridge, UK

Hassan Rivaz, Concordia University, Canada

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| 13:00-13:15 | BigFUSE: Global Context-Aware Image Fusion in Dual-View Light-Sheet Fluorescence Microscopy with Image Formation Prior
Speaker: Carsten Marr, Helmholtz Munich, Germany |
| 13:15-13:30 | Physics-based Decoding Improves Magnetic Resonance Fingerprinting
Speaker: Pingfan Song, University of Cambridge, United Kingdom |
| 13:30-13:45 | LLCaps: Learning to Illuminate Low-Light Capsule Endoscopy with Curved Wavelet Attention and Reverse Diffusion
Speaker: Long Bai, The Chinese University of Hong Kong, Hong Kong SAR, China |
| 13:45-14:00 | Physics-Informed Neural Networks for Tissue Elasticity Reconstruction in Magnetic Resonance Elastography
Speaker: Matthew Ragoza, University of Pittsburgh, USA |
| 14:00-14:15 | Learned Alternating Minimization Algorithm for Dual-Domain Sparse-View CT Reconstruction
Speaker: Chi Ding, University of Florida, USA |
| 14:15-14:30 | Inter-slice Consistency for Unpaired Low-Dose CT Denoising using Boosted Contrastive Learning
Speaker: Jie Jing, Sichuan University, China |